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MyDesign Portfolio

Element E Initial Template - AT - 15 points

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements (E1)

This section should describe how your proposed solution is supported by STEM (science, technology, engineering, mathematics, computer science, etc.) concepts.

1. Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.
2. Be sure to apply the applicable STEM concepts in context, directly to the specifics of your design. Review concepts learned in your other classes and tackle new concepts as appropriate. Include any relevant calculations.
3. Element C indicates that you should have at least three high-priority design requirements to receive a score of 3 (and four for a score of 4, and five for a 5). Your number of design requirements and your number of high-priority design requirements should be the same throughout your portfolio. You should use the same three to five design requirements here too, and apply **at least three of them** to STEM concepts to achieve a score of 5 on E1. Use as many copies of this sub-template as needed.

High-Priority Design Requirement #1 from Element C - Type of Ice Melt/De-icer The ice melts should not cause any additional cracking or be vulnerable to more damage over time to the concrete.

STEM Concepts Involved in the requirement: Material Science for Chemical Composition of Ice Melts/De-Icers

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Describe how STEM concepts apply directly to the specifics of your design requirement:

Material Science. De-icers should minimize damage to concrete to avoid corrosion

High-Priority Design Requirement #2 from Element C - Quality of Materials To ensure use of high quality materials during the process of repavement.

STEM Concepts Involved in the requirement: Environmental science. Materials need to be able to sustain the environment like having good weather and environmental resistance.

Describe how STEM concepts apply directly to the specifics of your design requirement:

Environmental Science, Life cycle assessment helps determine whether a material is sustainable.

High-Priority Design Requirement #3 from Element C - Proper Curing The concrete must be cured to allow the proper hydration, if the curing process is too fast then the concrete can dry too quickly and crack.

STEM Concepts Involved in the requirement: Technology. Moisture Control can help regulate curing like automated systems to monitor humidity.

Describe how STEM concepts apply directly to the specifics of your design requirement:

Technology. Temperature Monitoring can be used to track the temperature of the concrete



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High-Priority Design Requirement #4 from Element C (if you have one)- Control of Temperature and Weather. We can try to maintain a temperature of 20-25°C, which almost guarantees the durability and structural integrity of concrete surfaces.

STEM Concepts Involved in the requirement: Thermodynamics. Heat conduction causes the concrete surface to absorb or lose heat depending on the surrounding temperature, affecting their curing process.

Describe how STEM concepts apply directly to the specifics of your design requirement: Controlling the heat flow by using heat exchange methods like cooling jackets or temperature control systems.

High-Priority Design Requirement #5 from Element C (if you have one) - Control of Shrinkage

STEM Concepts Involved in the requirement: Chemistry. Chemical treatments or coatings can be applied to concrete to improve moisture retention and slow down the drying process which reduces shrinkage.

Describe how STEM concepts apply directly to the specifics of your design requirement: The rate of hydration can influence shrinkage. Fast hydration leads to rapid moisture loss, which leads to more shrinkage.



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Proposed Solution to STEM Principles and Practices (E2)

While E1 was focused on how STEM applies to the design requirements, E2 is focused on how STEM is used to support the design solution.

For E2, use STEM concepts to justify your design decisions. It's important that the STEM Principles and Practices that are discussed are actually appropriate and useful to the problem. Use as many copies of this sub-template as needed.

STEM Principles for Design Solution #1

STEM Concept: **Material Science (De-icers)**

Describe how STEM concept applies directly to the specifics of the first solution to your problem: **Concrete's reaction with de-icers depend on the permeability and structure of the concrete. De-icers on the concrete surface can cause cracking and scaling due to moisture absorption, accelerating, deterioration.**



You can only work on the next sections after you have built and tested your first prototype. You need to skip to the "Evidence is Reviewed" section.

STEM Principles for Design Solution #2

STEM Concept: **Technology (Proper curing)**

Describe how STEM concept applies directly to the specifics of your solution: **The use of technology in the curing process helps ensure that the hydration of concrete occurs optimally, preventing rapid moisture loss, cracking, and other issues associated with improper curing.**



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STEM Principles for Design Solution #3 (if completed)

STEM Concept: Thermodynamics (Control of Temperature)

Describe how the STEM concept applies directly to the specifics of your solution: The design requirement to maintain a temperature range of 20-25C for concrete curing and hardening directly involves thermodynamics, which governs the heat flow, energy transfer, and temperature regulation within materials.

**Evidence Design is Reviewed (E3)**

"Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

E3 is about experts reviewing what you've said about the STEM connections in E1 and E2. It's important here to talk to an appropriate field expert who knows about the STEM content of the project - not to a stakeholder.

Documentation #1 (Only one documentation section needed for full credit):

What format: Dr Butler helped us with our design by helping us find the proper items And materials for our design.



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- What the expert said about the connection you've made from your design requirements to STEM principles and practices:
- What the expert said about how STEM principles and practices substantiate your design solution:

Documentation #2 (+ making only):

What format: Interview notes, emails, audio/video tape with transcript, etc.

- What the expert said about the connection you've made from your design requirements to STEM principles and practices:
- What the expert said about how STEM principles and practices substantiate your design solution:

Documentation #3 (+ making only):

What format: Interview notes, emails, audio/video tape with transcript, etc.

- What the expert said about the connection you've made from your design requirements to STEM principles and practices:
- What the expert said about how STEM principles and practices substantiate your design solution:

Experts with Credentials (E4)

Document who your field experts are (names are not necessary) and what qualifies them as an expert. Use as many copies of this sub-template as needed.



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Expert #1 (from documentation 1): Dr Butler is a renowned engineer who has collaborated with many designers and professionals throughout the years

Expert #2 (from documentation 2): Describe credential(s) and what makes them qualified as an expert.

Expert #3 (from documentation 3): Describe credential(s) and what makes them qualified as an expert.

Verification or Corrective Response to Design Review (E5)

Your conversation/interview with the expert(s) should inform some corrective responses. In E5, describe how those reviews informed your responses.

For example,

"_____ told us _____, which informed our decision to change/reconsider _____." (or "...which verified/reinforced our decision to _____.")

Make sure to document the experts' reviews (suggestions/feedback or criticisms of your proposed solution) and how they will inform iterations you will make to your proposed solution. Use as many copies of this sub-template as needed.

Interview with Expert #1 informs corrective response:

Interview with Expert #2 informs corrective response:

Interview with Expert #3 informs corrective response:

Criteria	5	4	3	2	1	0

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3. Evidence that Design is Reviewed. There is _____ evidence that the application of STEM principles & practices (or suitable alternative) has been reviewed by expert(s).	<i>substantial</i>	<i>adequate</i>	<i>some</i>	<i>minimal</i>	<i>no - but review was requested -</i>	<i>no</i>
4.Design Reviewed by Experts. _____ expert(s) have/has reviewed the design for application of STEM principles & practices.	<i>Two or more, with at least two with documented, relevant credentials</i>	<i>Two or more, with at least one with documented, relevant credentials</i>	<i>At least one with documented, clearly relevant credentials</i>	<i>At least one with documented credentials</i>	<i>Only one with unclear credentials</i>	<i>No</i>
5.Design Review Results in Verification. The design reviews provided _____.	<i>clear and detailed confirmation (verification) or substantial detail necessary to inform a corrective response</i>	<i>clear confirmation (verification) or detail necessary to inform a corrective response</i>	<i>partial confirmation (verification) or some detail necessary to inform a corrective response</i>	<i>does not provide clear confirmation (verification) or suggestions for a corrective response, but at least some response is provided</i>	<i>show no evidence of confirmation (verification) or suggestions for a corrective response by an expert, but includes notes of attempts to gain feedback</i>	<i>show no evidence of confirmation (verification) or suggestions for a corrective response by an expert</i>



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